HANDOUT – 4
COMPUTATIONAL THINKING
Topic: Directions and Coordinates

ARAVALI INTERNATIONAL SCHOOL

Date: ___________

Name: ______________________ Class: ___________ Sec: ___________ Roll No.: ____________

1.2 Directions and Coordinates

Suppose your friend asks, "Where is the school library?" You could give directions "two corridors right, then up one floor" or a coordinate "at (4, 3)". Both describe a location precisely—similar to how computers, robots and drones locate objects.

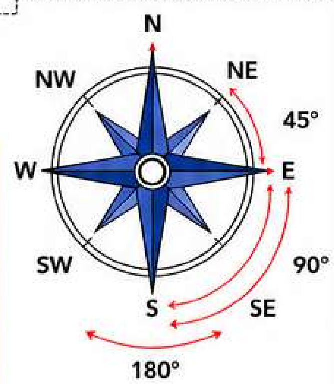
The Compass

The four main directions are North, South, East, and West. We also use four in-between directions in real life to describe locations more accurately—NE (North-East), NW (North-West), SE (South-East), SW (South-West).

Turning by angles

A change in direction can be measured by an angle. For example, Quarter-turn = 90°; Half-turn = 180°; Turn from N to NE = 45°.

> Computers love angles. Every time a robot in a factory or a Mars rover changes direction, it does so by a precise number of degrees — never more, never less.



Activity 1 CRICKET PRACTICE AND TEMPERATURE

Coach Meera is planning a summer cricket camp and wants to explore how temperature affects attendance at outdoor practice across different Indian cities.
Look at the scatter plot and answer the following questions.

1. Which city has the highest attendance?

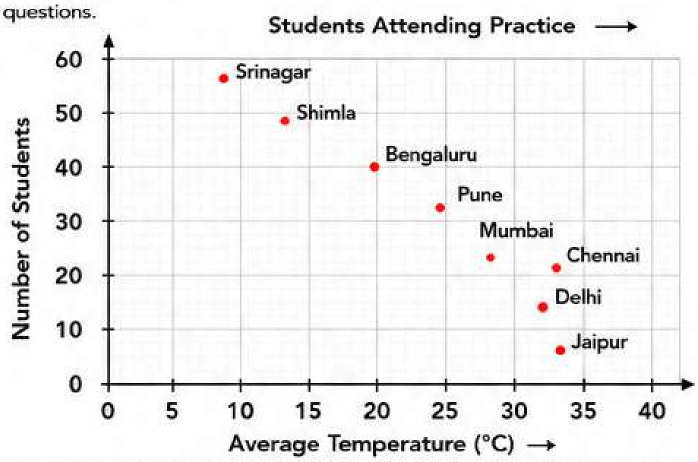
   ____

2. What happens to attendance as temperature increases?

   ____

3. Is there a positive or negative relationship?

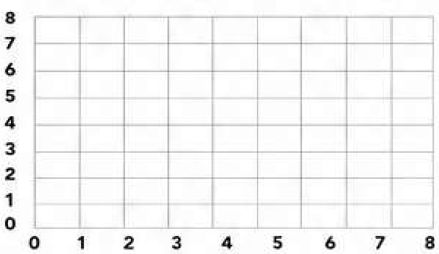
   ____



Activity 2 DRONE DELIVERY

A drone network delivers medicines to hospitals across India. The drone's navigation system does not understand location names—it only reads grid coordinates. Before every flight, a database operator must plot each hospital on the grid and plan the most efficient route.

1. Plot the following hospitals on the grid.



Hospital Code	Grid Coordinates
H-01	(4, 6)
H-02	(3, 5)
H-03	(4, 4)
H-04	(2, 3)
H-05	(5, 3)
H-06	(4, 1)
H-07	(7, 5)

2. The drone is currently at H-01. It must deliver medicines to H-02 and H-05. The battery is limited and will drain out if it takes a longer route. Which hospital should it deliver to first?

   a) H-02

   b) H-05

   c) Both are equally far

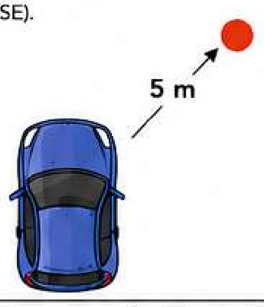
   d) Cannot be determined

Activity 3 SELF-DRIVING CAR

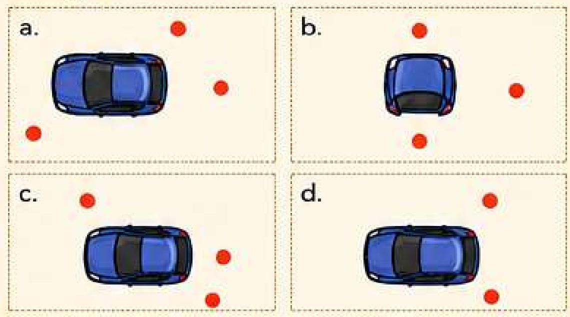
A self-driving car is being tested. It can detect objects around it. It has been designed to send the location of the objects including two data points.

1. The direction (N, W, E, S, NW, SW, NE, SE).
2. Distance in metres from the car.

The directions are calculated from the front of the car. For example, if the car sends a message (NE, 5), it would mean there is an object at 5m in the NE direction.



The car sends the message (NE, 5) (S, 10) (SW, 5). Which of the cars given below is sending this message:



Curriculum Connection - Maths (Connecting the Dots) . S.Sc. (Understanding the Weather)